exp-1

```
package main
import "fmt"
func main() {
  var a, b, x, y, rem int
  fmt.Scan(&a, &b)
  x, y = a, b
  for y != 0 {
  rem = x % y
  x = y
  y = rem
 }
 gcd := x
 lcm := (a * b) / gcd
 fmt.Println("GCD =", gcd)
 fmt.Println("LCM =", lcm)
}
```

exp-2

```
package main
import "fmt"
func main() {
 var n int
 fmt.Scan(&n)
 for i := 1; i <= n; i++ {
  for j := 1; j <= n-i; j++ {
   fmt.Print(" ")
  }
  for j := 1; j <= i; j++ {
   fmt.Print(j, " ")
 }
 if i>1{
     fmt.Println(1)
 }
 fmt.Println()
}
}
```

exp-3

```
package main
import "fmt"
func Square(x int)int{
   return x*x
}
func main() {
 result:=Square(10)
 fmt.Println(“square of x:”,result)
}
```

exp-4

```
package main
import (
            "fmt"
            “math”
 )
func main() {
  var sum,mean,sd float64
  num=[5]float64{10,20,30,40,50}
  for i:=0;i<5;i++{
       sum+=num[i]
  }
  mean=sum/5
  for i:=0;i<5;i++{
    sd+=math.Pow(num[i]-mean,2)
  }
  sd=math.sqrt(sd/5)
 fmt.Println(“standard deviation:”,sd)
 fmt.Println(“Mean:”,mean)
}
```

exp-5

```
package main
import "fmt"
func main() {
  var num,rows int
  fmt.Print(“enter row number”)
  fmt.Scan(&rows)
  num=1
  for i:=0;i<=rows;i++{
    for j:=0;j<=i;j++{
      fmt.Print(num,” ”)
      num++
    }
  }
 fmt.Println(“\n”)
}
```

exp-6

```
package main
import "fmt"
func main() {
  var str1,str2 int
  fmt.Print(“enter string 1 :”)
  fmt.Scan(&str1)
  fmt.Print(“enter string 2 :”)
  fmt.Scan(&str2)
  result=str1+str2
  fmt.Println(“Concatenation string :”,result)
}
```

exp-7

```
package main
import "fmt"
func main() {
  var str int
  fmt.Print(“enter string 1 :”)
  fmt.Scan(&str)
  flag:=true
  for i:=0;i<len(str)/2;i++{
    if str[i]!=str[len(str)-1-i]{
        flag:=false
        break
    }
  }
  if flag{
     fmt.Print(“Palindrome”)
  else{
     fmt.Print(“Not Palindrome”)
  }
}
```

exp-8

```
  package main
  import "fmt"
  type contact struct(){
          Name string
          email string
          message string
  }
  func main() {
    var c contact
    fmt.Print(“enter Name :”)
    fmt.Scan(&c.Name)
    fmt.Print(“enter Email :”)
    fmt.Scan(&c.email)
    fmt.Print(“enter Message :”)
    fmt.Scan(&c.message)
    fmt.Print(“Contact Details”)
    fmt.Println(“Name:”,c.Name)
    fmt.Println(“Email:”,c.email)
    fmt.Println(“Message:”,c.message)
    fmt.Print(“Contact Saved”)
  }
```

exp-9

```
package main
import "fmt"
func main() {
  var n ,sum int
  fmt.Print(“enter N number :”)
  fmt.Scan(&n)
  arr:=make([]int,n)
  for i:=0;i<n;i++{
     fmt.Scan(&arr[i])
     sum+=arr[i]
  }
  avg:=float64(sum)/float64(n)
  fmt.Println(“Average:”,avg)
}
```

exp-10

```
package main
import "fmt"
func main() {
  arr:=[]int{1,2,3,2,4}
  result:=[]int{}
  seen:=make(map[int]bool)
  for i:=0;i<len(arr);i++{
     if !seen[arr[i]]{
        seen[arr[i]]=true
        result=append(result,arr[i])
     }
   }
   fmt.Println(“Original Array :”,arr)
   fmt.Println(Removed duplicate Array:”,result)
}
```

exp-11

```
package main
import "fmt"
func main() {
  num:=[]int{1,2,5,8,3}
  fmt.Println(“original Integer:”,num)
  sort.Sort(sort.Reverse(sort.IntSlice(num)))
  fmt.Println(“Sorted Integer:”,num)
  str:=[]string{”Anu”,”Ravi”,”Shiva”,”Neha”}
  fmt.Println(“Original String:”,str)
  sort.Sort(sort.Reverse(sort.StringSlice(str)))
  fmt.Println(“Sorted string:”,str)
}
```

exp-12

```
package main
import "fmt"
func main() {
  str:= “Go Programming”
  fmt.Println(“Contains:”,strings.Contains(str,”Go”)
  fmt.Println(“Contains Any:”,strings.ContainAny(str,”xyzGO”))
  fmt.Println(“Count:”,strings.Count(str,”o”))
  fmt.Println(“Equal Fold:”,strings.EqualFold(“Hello”,”HELLO”)
}
```